1. Current approach and changes since M1

Architectural Paradigm & Core Approach

TOSS / Life departs fundamentally from passive, text-heavy financial literacy curricula by placing youth (ages 18–35) into an active, choice-consequence lifecycle simulation spanning 5 sequential developmental chapters. Engineered as an offline-first monorepo (Next.js 16, React 19, Tailwind CSS v4, PGlite, and hosted Supabase backend), the platform delivers mathematically sound financial exposure. Every turn models realistic economic forces: compounding inflation eroding idle cash, formal banking vs. predatory lenders, risk mitigation (emergency funds and health cover), and social peer dynamics.

CLIENT TIER: Offline-First Next.js 16 Progressive Web App (PWA)
React 19 • Tailwind CSS v4 • Cache-First Service Worker (sw.js) • Indian Vernacular Formats (₹ Lakh/Crore)

Deterministic Engine & Content Packages
- Multi-stream PRNG: Seeded determinism for "What-If" replays
- 5 Narrative Chapters: Strict schema validation (Zod & Ajv)
- Life Ladder & Compounding: Real inflation vs. nominal cash

Hosted Supabase Cloud & Multi-Tenant Analytics
- Live Friends' Room: Authoritative lockstep turn execution via RPC
- Collaborative Finance: Chit fund auctions, loans, dynamic trust
- NGO Dashboard: Cohort metrics, seat redemption, pre/post delta

Major Features & Capabilities Shipped Since M1

All 5 Narrative Chapters Implemented & Tuned
Full lifecycle trajectory: Ch1 (Budgeting & emergency reserve), Ch2 (Career, formal debt vs. loan sharks, health shocks), Ch3A/Ch3B (Entrepreneurship, working capital, mutual funds), and Ch4 (Inflation compounding, bear/bull markets, retirement).

Friends' Room Multiplayer & Trust Mechanics
Shipped live server-authoritative multiplayer backed by Supabase Edge RPCs. Supports rotating chit funds, peer emergency lending, pooled risk bidding, and dynamic trust scoring updated upon loan repayment.

Diagnostic Money Mindset Checks
Pre- and post-chapter psychometric evaluations quantifying behavioral traits: delayed gratification, risk composure, and financial locus of control—mapping players into personalized pedagogical personas.

Counterfactual "What-If" Replay Engine
Post-chapter diagnostic powered by deterministic seeded state branching. Players selectively alter past choices (e.g., "What if I purchased health insurance in Year 2?") to inspect diverging wealth curves.

Telugu Vernacular Localization Parity
Achieved 100% key parity (246/246 keys in te.json) with Indian financial conventions, vernacular idioms, and translated investment options for Tier-2/3 student comprehension.

Offline-First Resilience & PWA Support
Engineered comprehensive service worker asset caching (sw.js) and embedded client PGlite Wasm database, guaranteeing uninterrupted gameplay in bandwidth-constrained NGO classrooms.

2. Experiments run so far and their results (Part 1)

A. Headless Monte Carlo Simulation Balancing (400 Full Runs Across Chapters)

To verify that the engine rewards disciplined financial habits without becoming punitive or trivial, we executed automated Monte Carlo simulations across three archetype strategies: *Balanced*, *All-Cash*, and *Impulsive*.

Strategy Archetype	Ch1 Pass	Ch2 Pass	Ch3A Pass	Ch3B Pass	Ch4 Pass	Pedagogical Finding & Failure Modes
Balanced / Disciplined	**87.2%**	**98.0%**	**100.0%**	**95.1%**	**100.0%**	Withstands random health shocks; achieves sustainable growth.
All-Cash / Ultra-Conservative	94.0%	96.0%	72.0%	61.5%	**0.0%**	**Exposes Cash Illusion: Inflation fully destroys terminal purchasing power.**
Impulsive / Speculative	**42.5%**	**67.0%**	**38.0%**	**45.0%**	**14.0%**	Falls into predatory debt spirals and scam investments.

2. Experiments run so far and their results (Cont.)

B. Engine Latency on Low-End Mobile (pnpm bench)

Experiment: Benchmarked engine state evaluations over 1,000 multi-asset, event-heavy turns under 4x CPU throttling.
Result: Resolves in **0.09 ms** natively and **< 0.4 ms** on budget mobile hardware (well within the 50 ms budget).

C. Seeded PRNG Determinism & What-If Branches

Experiment: Verified 100,000 state mutations across identical random seeds in rng.test.ts.
Result: **100% deterministic reproducibility**. Enables accurate counterfactual comparisons when players modify past decisions.

D. Concurrency & Network Jitter in Multiplayer Rooms

Experiment: Injected synthetic 1,500 ms packet latency across concurrent chit fund bidding and peer lending rounds.
Result: Authoritative Postgres RPC row locks resolved bids sequentially with zero state desync or double-crediting anomalies.

E. Local Offline Hotspot & LAN Testing

Experiment: Evaluated pnpm dev:lan over unencrypted HTTP LAN without active internet.
Result: Fallback UUID generation and clipboard APIs functioned reliably, proving offline viability in rural educational setups.

3. Progress on testing

We enforce four continuous verification quality gates prior to merging code, ensuring total integrity across the simulation engine, database, and front-end user experience:

Test Layer & Harness	Coverage & Scope Description	Tests	Duration	Verification Status
Engine & Mechanics (Vitest)	Compound inflation, debt compounding, Life Ladder, scam mechanics, Money Mindset, What-If counterfactuals.	**119 tests**	~2.1s	**100% Passing**
Supabase & Database (PGlite)	Atomic student code redemptions, room server concurrency, multi-tenant NGO seat view security.	**27 tests**	~7.8s	**100% Passing**
Web & Formatters (Vitest)	Multiplayer room backends, Indian currency formatters (Lakh/Crore), session state serialization.	**11 tests**	~0.2s	**100% Passing**
Content Validation (Zod / Ajv)	21 YAML files: Probability weights sum to 1.0, investment yields within ±0.5 pp, schema integrity.	**21 files**	~1.5s	**Zero Errors**
E2E Mobile Journeys (Playwright)	Pixel 5 mobile viewport: Full playthrough, scam interception modal, NGO teacher dashboard.	**4 suites**	~14s	**All Suites Pass**
TypeScript & Lint Quality	Root monorepo, Next.js web application, shared packages (engine, content, db-types).	**Monorepo**	~3.2s	**0 Errors / 0 Warnings**

4. Plan for the remaining time

With the core simulation engine, narrative chapters, and test pipeline fully stabilized, our remaining hours are dedicated to production hardening, localized content expansion, and showcase rehearsal:

STEP 1
Deploy & Harden
Vercel Edge + Supabase Prod; lock down RPC execute privileges.

STEP 2
Content Parity
Extend Telugu translations to Ch1/3/4; Hindi & Urdu starter lexicons.

STEP 3
Mentor Review
Calibrate yield and penalty bounds against real Indian market stats.

STEP 4
Final Showcase
Rehearse 6-step judges' evaluation rubric & live NGO cohort demo.

- **Production Cloud Deployment & Security Hardening:** Finalize Vercel edge deployment wired to hosted Supabase (`vapzfepyyjdsxxjclpwr`). Audit Row Level Security policies and restrict execute privileges on atomic code redemption RPCs (`redeem_student_code_claim`) to authenticated service roles.
- **Vernacular Expansion & Content Localization:** Extend Telugu narrative translations beyond Chapter 2 and investment items into Chapters 1, 3, and 4. Add initial Hindi and Urdu glossary mappings for financial vocabulary.
- **Mentor Calibration & Threshold Fine-Tuning:** Review simulation investment yield bounds (e.g. index funds vs. chit funds) with Vanguard mentors to align game mechanics with real-world Indian market indicators.
- **End-to-End Live Rehearsal:** Practice presentation against the 6-step judges' evaluation rubric: Diagnostic Mindset Check → Turn Simulation → Scam Interception → Multiplayer Friends' Room → What-If Counterfactual Report → NGO Cohort Analytics.